

Houdaifa Drahm

Backend & DevOps Engineer

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SUMMARY

Backend & DevOps Engineer specialized in building high-performance Node.js/NestJS microservices and scalable containerized infrastructure. Expert at bridging low-level systems programming (C/C++, POSIX) with modern DevOps automation including CI/CD pipelines, Docker orchestration, and real-time WebSocket architectures. Currently completing the 42 Network engineering curriculum at 1337 School (UM6P).

TECHNICAL SKILLS

- **Languages & Tools:** C, C++, JavaScript (ES2022+), TypeScript, Python, Bash, Git, Docker, Kubernetes, Make, Vim, Valgrind, GDB
- **Backend & Networking:** Node.js, NestJS, REST APIs, WebSockets, Socket.io, JWT Auth, HTTP/1.1, TCP/UDP sockets, Multithreading, Concurrency
- **DevOps & Infrastructure:** Docker Compose, GitHub Actions CI/CD, DigitalOcean, Nginx (TLS/SSL), Linux & Windows Server, System Administration, Virtualization, Monitoring
- **Systems Programming:** POSIX Threads, Mutexes, Semaphores, Signal Handling, Process & Memory Management, I/O Multiplexing: poll()/select()
- **Databases:** PostgreSQL, MariaDB

PROJECT EXPERIENCE

- **ft_transcendence:** Engineered a full-stack real-time multiplayer platform using Node.js, NestJS, and Socket.io. Implemented JWT-based authentication with refresh-token rotation, WebSocket game rooms, and live chat. Automated deployments to DigitalOcean via GitHub Actions CI/CD pipeline with zero-downtime containerized releases.
- **Webserve:** Built a production-grade HTTP/1.1 server from scratch in C++ with no external libraries. Non-blocking I/O via poll()/select() handles concurrent connections; supports GET, POST, DELETE, chunked transfer encoding, and CGI execution.
- **Inception:** Architected a multi-container infrastructure using Docker Compose: Nginx with TLS termination, MariaDB with persistent volumes, and WordPress in isolated containers with custom networking.
- **Minishell:** Implemented a fully functional Unix shell in C with Bash-compliant parsing and execution. Supports multi-pipe chaining, heredocs, I/O redirections, environment variable expansion, and signal handling (SIGINT/SIGQUIT). 100% memory-safe confirmed via Valgrind.
- **Philosophers:** Concurrent dining philosophers simulation using pthreads and mutexes. Guarantees deadlock-free and starvation-free execution. Zero data races under stress testing.
- **Core 42 Projects:** Libft, ft_printf, Get_Next_Line, Born2beroot (Linux VM hardening), NetPractice (TCP/IP subnetting & routing), CPP Modules 00–09 (OOP, templates, STL, polymorphism, exceptions).

EDUCATION

1337 School (42 Network), UM6P, Bengueurir 2023 – **Present**
Common Core Program in Systems & Advanced Programming, peer-reviewed, project-based engineering curriculum.

ISTA, Beni Mellal 2021 – 2023
Diploma in Digital Infrastructure, Networking & System Administration

ACHIEVEMENTS

- **Architected** and shipped a production-level real-time multiplayer backend with WebSocket game rooms, JWT authentication, and automated CI/CD pipeline deploying to DigitalOcean, built entirely from scratch with zero-downtime containerized releases.
- **Automated** CI/CD pipelines with GitHub Actions targeting DigitalOcean across containerized services, significantly improving deployment speed and achieving full environment reproducibility on every release.
- **Engineered** an HTTP/1.1 server in C++ capable of handling concurrent connections through non-blocking I/O multiplexing, with no external libraries, full CGI support, and chunked transfer encoding.

LANGUAGES

Arabic *Native* **English** *Professional* **French** *Professional*